

NeuroForge Nexus

Cloud-Native Software Development Lifecycle Management Platform

8 Weeks | 4 Milestones

1. Project Idea

NeuroForge Nexus is an enterprise-grade platform that manages the complete Software Development Lifecycle (SDLC) from requirements gathering to deployment, monitoring, and release management. The platform enables organizations to manage Requirements, Projects, Tasks, Sprint Planning, Code Repositories, CI/CD Pipelines, Testing, Release Management, Deployment Tracking, and DevOps Monitoring. The platform uses Angular 20 frontend with Java 25 and Spring Boot 4 backend. PostgreSQL database with Redis cache. Microservices include User Service, Project Service, Sprint Service, CI/CD Service, Release Service, and Reporting Service. Kafka enables event-driven architecture. Keycloak provides IAM. The system manages 247 active projects, 2,847 users across 47 teams, processes 1,247 builds daily with 97.4% success, and maintains 99.99% uptime.

2. Emerging Skills

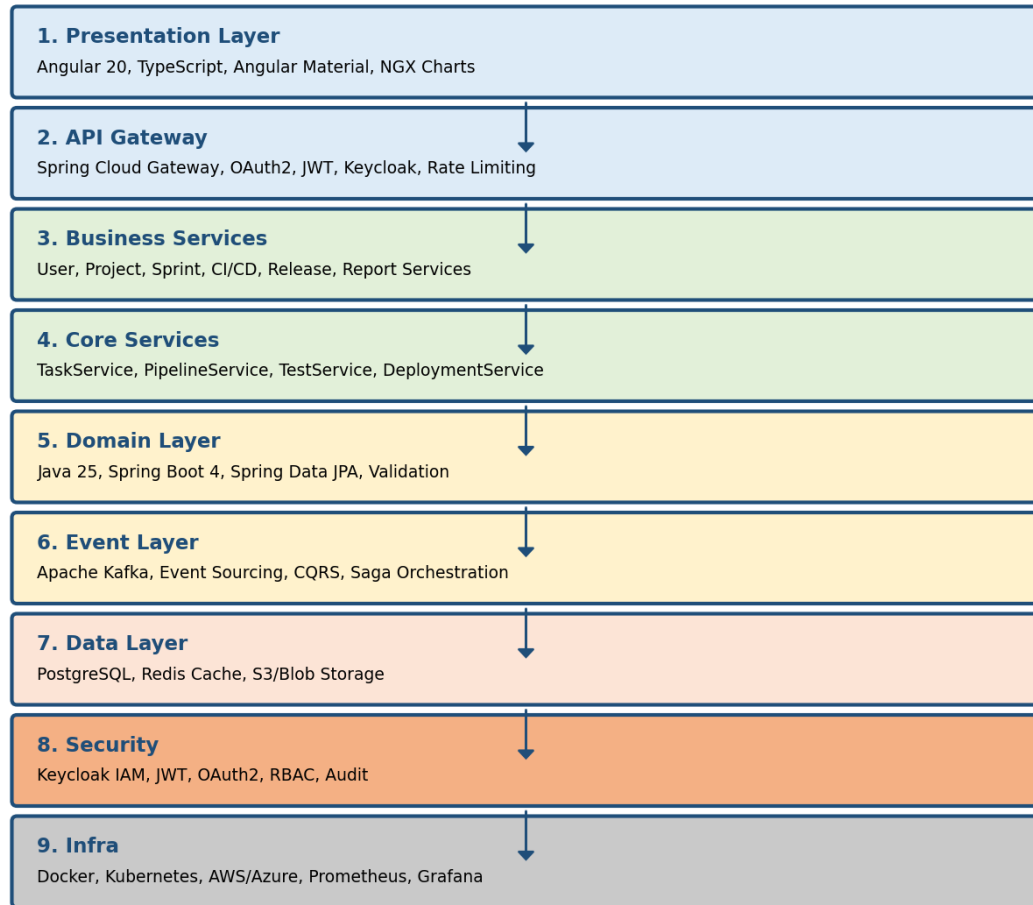
- Full Stack Java, Cloud, DevOps, Microservices, Event Driven Architecture, Banking Domain, Security

3. Technology Stack

Component	Technology
Frontend	React / Angular 20+
Backend	Java 25, Spring Boot 4
Database	PostgreSQL / MySQL
Cloud	AWS EC2, RDS, S3, or Azure App Service, SQL, Blob
Skills	Full Stack Java, Cloud, DevOps, Microservices, Event Driven Architecture, Banking Domain, Security

4. System Architecture Diagrams

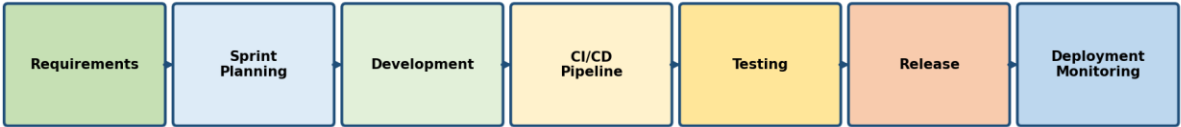
NeuroForge Nexus - 9-Layer SDLC Microservices Architecture



9-layer architecture with Angular 20 frontend, Spring Cloud Gateway, Java 25/Spring Boot 4 microservices, Kafka event streaming, and PostgreSQL data layer.

5. Project Working Diagram

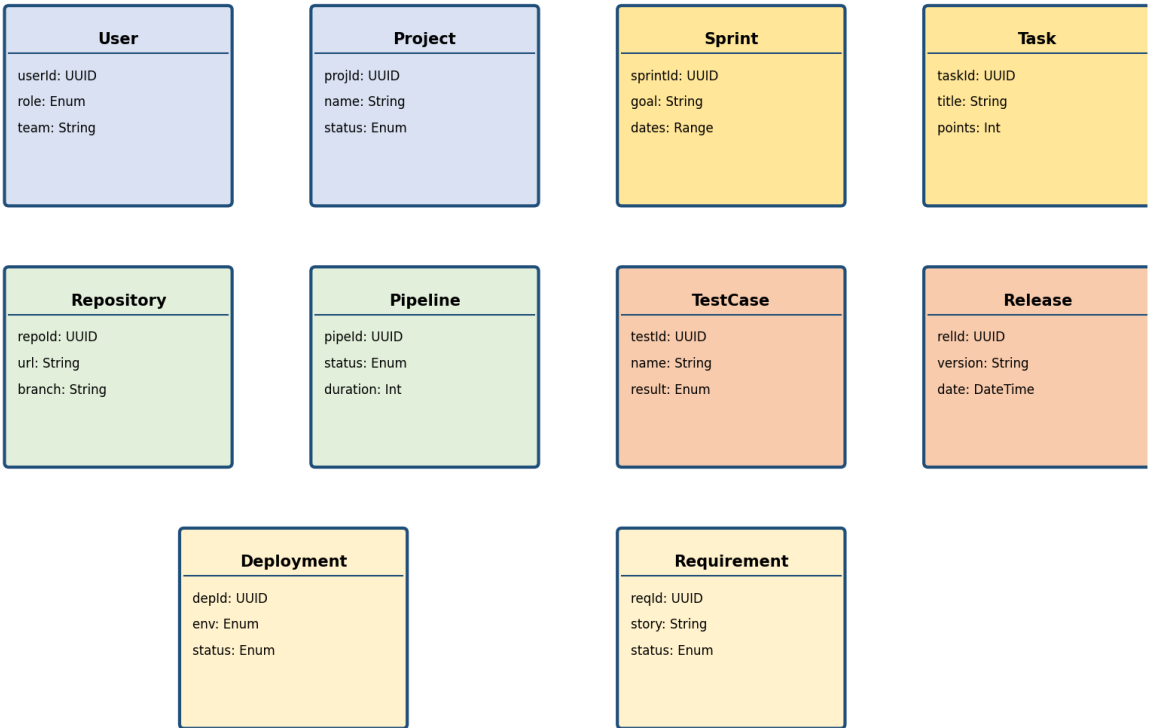
NeuroForge Nexus - Complete SDLC Workflow



Complete SDLC flow: Requirements → Sprint Planning → Development → CI/CD Pipeline → Testing → Release → Deployment Monitoring.

6. Class Diagrams

NeuroForge Nexus - Class Diagram: SDLC Domain



Core entities: User, Project, Sprint, Task, Repository, Pipeline, TestCase, Release, Deployment, Requirement.

7. Milestone-wise Deliverables & Screens

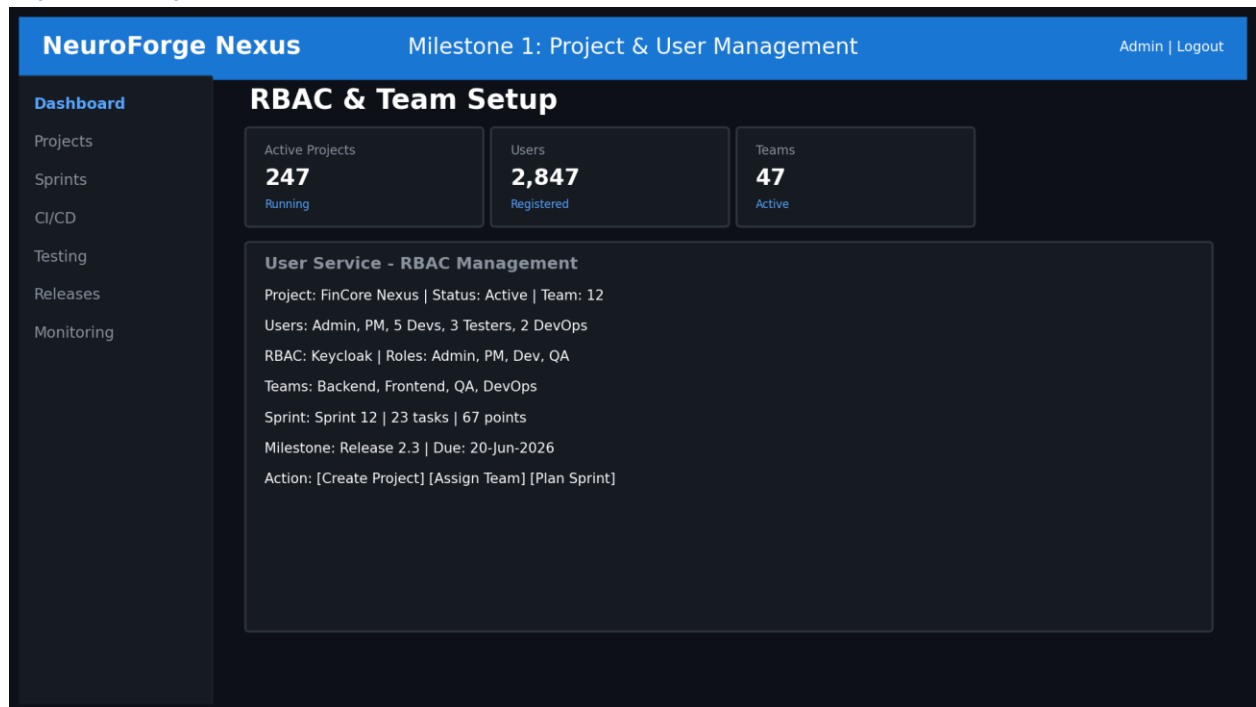
Milestone 1 (Weeks 1-2) – Project & User Management

Implements user registration, login, RBAC, and team management. Manages 247 projects with 2,847 users. Supports roles: Admin, Project Manager, Developer, Tester, DevOps Engineer. Integrates Keycloak for IAM. Enables sprint planning and milestone tracking.

Key Modules

- User Service
- Project Service
- Team Management
- RBAC
- Sprint Planning
- Milestone Tracking

Expected Output Screen



Project Service: FinCore Nexus project. Team: 12 members. Sprint 12 is active with 23 tasks. Milestone: Release 2.3 due 20-Jun-2026.

Validation Screens

- User registration
- RBAC enforcement

- Team assignment
- Project creation
- Sprint planning
- Milestone tracking

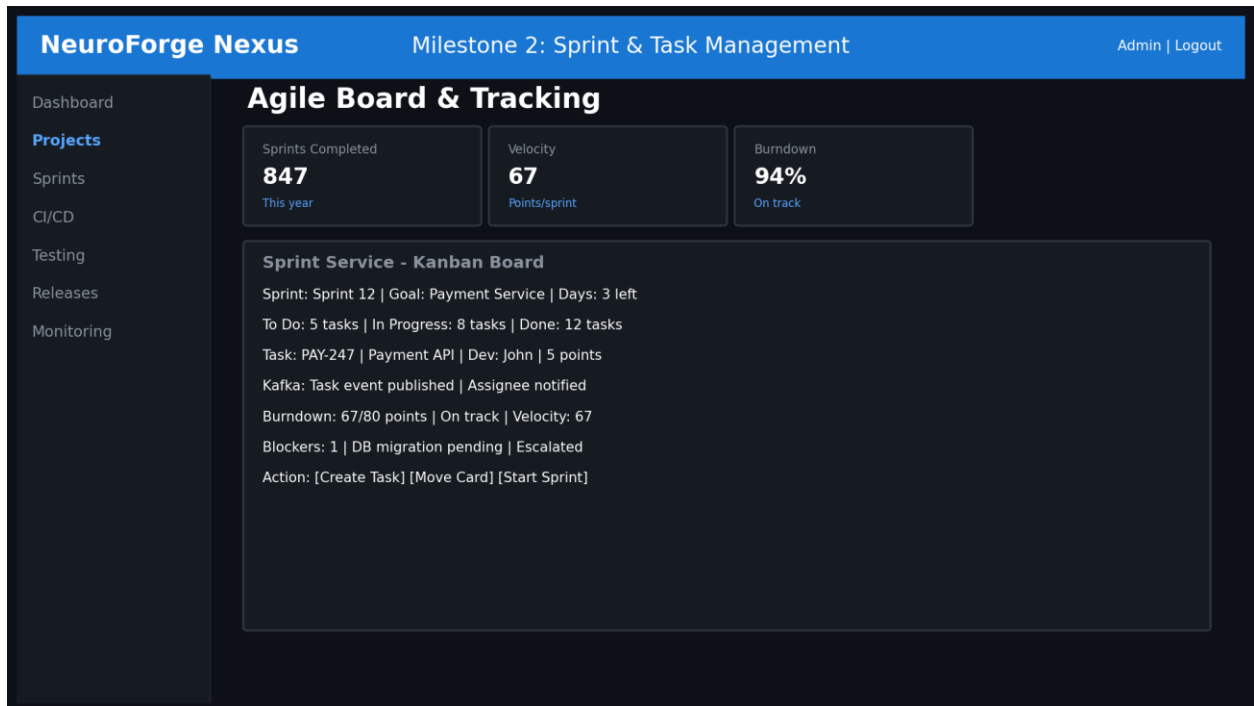
Milestone 2 (Weeks 3-4) – Sprint & Task Management

Deploys Kanban and Sprint boards. Completed 847 sprints this year. Maintains a velocity of 67 points. Achieves 94% burndown. Tracks task assignment and story points. Publishes task events to Kafka. Manages blockers and dependencies.

Key Modules

- Sprint Service
- Kanban Board
- Task Management
- Story Points
- Progress Tracking
- Kafka Events

Expected Output Screen



Sprint Service: Sprint 12. Goal: Payment Service. 5 To Do, 8 In Progress, 12 Done. Velocity: 67. Burndown: 67/80 points.

Validation Screens

- Sprint creation

- Task assignment
- Kanban board
- Burndown chart
- Velocity tracking
- Blocker management

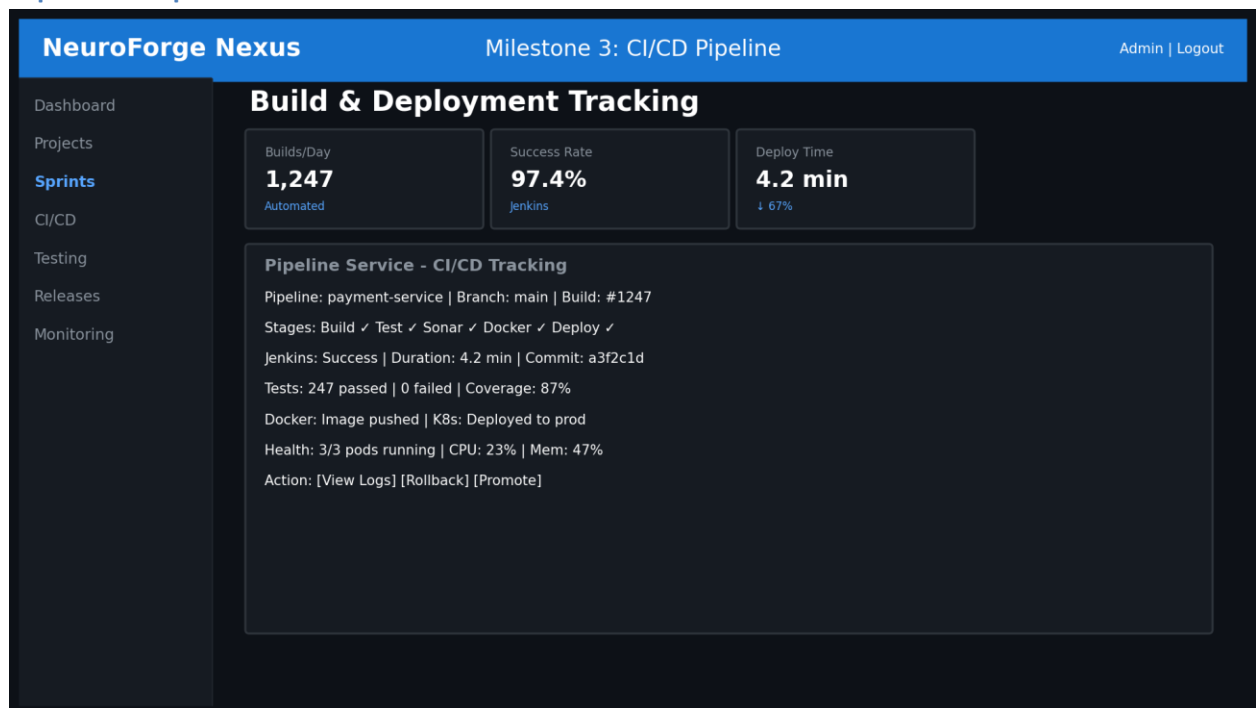
Milestone 3 (Weeks 5-6) – CI/CD Pipeline

Implements build and deployment tracking. Processes 1,247 builds daily. Achieves 97.4% success rate. Deploys in 4.2 minutes on average. Integrates Jenkins and GitHub Actions. Monitors pipeline health. Supports rollback management.

Key Modules

- Pipeline Service
- Build Tracking
- Deployment Tracking
- Jenkins Integration
- GitHub Actions
- Release History

Expected Output Screen



Pipeline Service: payment-service build #1247. Stages: Build ✓ Test ✓ Sonar ✓ Docker ✓ Deploy ✓. Success in 4.2 min.

Validation Screens

- Build tracking
- Pipeline monitoring
- Test results
- Deployment status
- Rollback capability
- Success rate

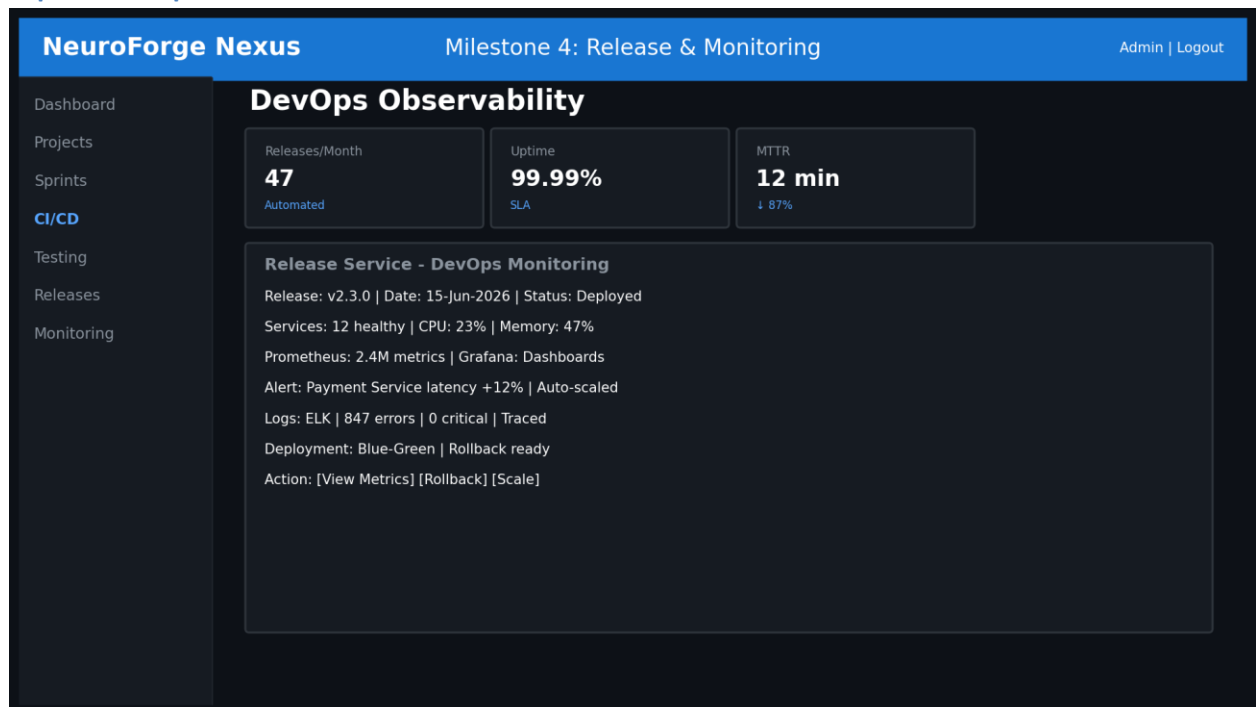
Milestone 4 (Weeks 7-8) – Release & Monitoring

Deploys DevOps monitoring dashboard. Manages 47 releases monthly. Maintains 99.99% uptime. Reduces MTTR to 12 minutes. Monitors service health, CPU, memory, and logs. Supports blue-green deployments.

Key Modules

- Release Service
- Deployment Tracking
- DevOps Monitoring
- Prometheus
- Grafana
- ELK Stack

Expected Output Screen

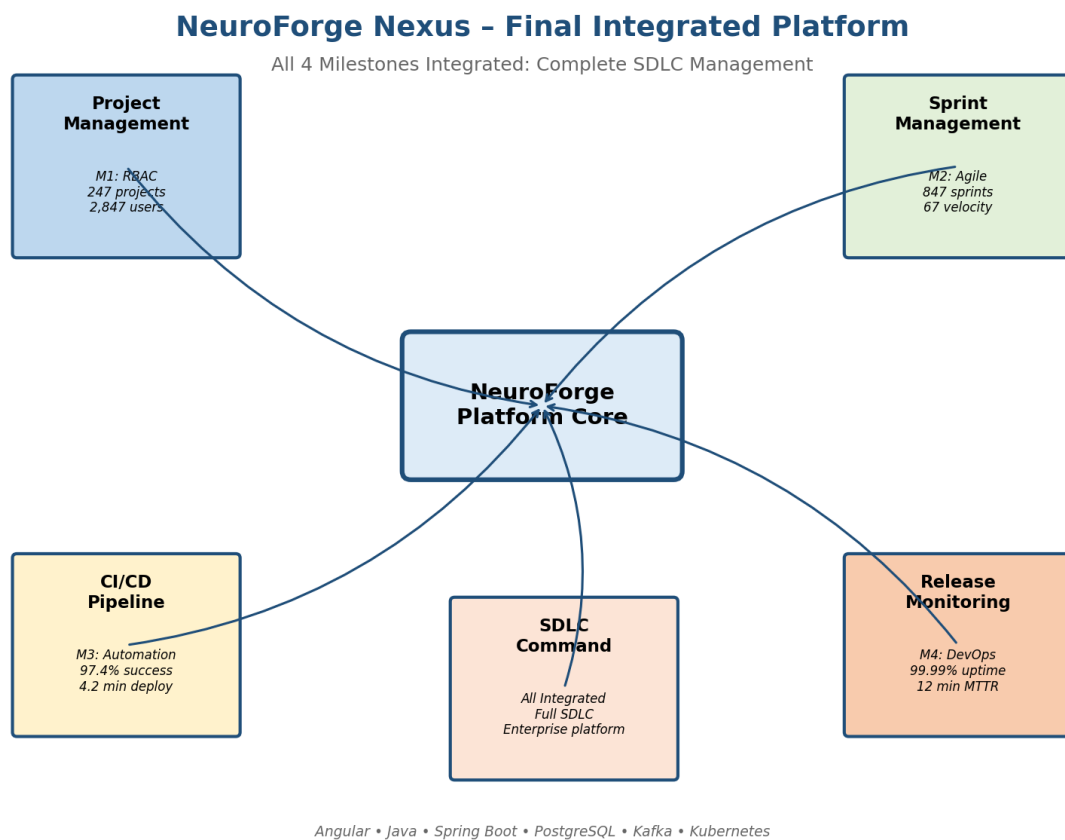


Release Service: v2.3.0 deployed. 12 services healthy. CPU 23%, Memory 47%. Alert: Payment Service latency +12%, auto-scaled.

Validation Screens

- Service health
- Resource monitoring
- Deployment tracking
- Alert management
- Log analysis
- Rollback validation

8. Master Application Screens – Final Integrated



Final integrated master screens after all 4 milestones:

- Project Management: M1 – RBAC, 247 projects, 2,847 users, 47 teams
- Sprint Management: M2 – Agile, 847 sprints, 67 velocity, 94% burndown
- CI/CD Pipeline: M3 – Automation, 97.4% success, 4.2 min deploy, 1,247 builds/day
- Release Monitoring: M4 – DevOps, 99.99% uptime, 12 min MTTR, 47 releases/month
- SDLC Command: All integrated – Full SDLC, enterprise platform, cloud-native

9. Validation Screens Framework

All screens enforce: User authentication via Keycloak, RBAC enforcement, Sprint velocity tracking, Pipeline success monitoring, Release compliance, and comprehensive audit logging for all SDLC actions. Operations show build status + deployment health + sprint progress. Critical releases require approval. Kafka ensures event ordering. All actions are logged immutably. Projects validated. Pipelines monitored. Releases audited.